

Alexandre Dalban

Citizenship: USA, EU | **Settled Status:** UK

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Education

CentraleSupélec, *MSc in Artificial Intelligence (M2)*

Paris, FR | Sept 2025 – Nov 2026

- Electives: AI for Finance, Stochastic Optimization, Reinforcement Learning, Advanced Deep Learning, Ensemble Learning.

Boston College, *BS in Mathematics and Computer Science*

Boston, USA | Aug 2021 – May 2025

- Mathematics: Probability, Statistics, Learning Theory, Mathematical Logic. CS: Algorithms, Machine Learning, Computer Systems.

Experience

Quantitative Finance Intern, *LeCap Asset Management (Hedge Fund)*

London, UK | Summers 2024, 2025

- Researched and validated 100+ alpha signals for systematic equity portfolio strategy, supporting portfolio construction.
- Built production Python tooling for the live trading environment, including an automated weekly reporting system evaluating 150+ signals across 8 performance metrics, replacing manual analysis.
- Built supervised forecasting pipeline spanning classical ML through transformer architectures, training on 90 proprietary indicators and Axioma factor-residualized returns across the top 1000 North American equities; best model achieved 54% daily directional accuracy vs. ~50% coin-flip baseline.
- Developed PPO reinforcement learning agents with reward based on rolling 5/10/20-day realized Sharpe and P&L; agents output dynamic indicator weightings producing a composite prediction signal.

Applied AI Research Intern, *Siemens AG*

Munich, GER | May 2026 – Oct 2026

- Worked on automating a manual manufacturing step: finding which surfaces of a 3D part design form each feature to be machined. Took the team's model (XGBoost) from 51% to 76% of features exactly right on real parts, with one-change, cross-validated experiments tracing the gain to cleaner labels (+14 points), more data (+11) and new features (+1).
- Built an LLM agent that recognises features with no training on the task, querying each part's exact geometry through 21 tools: 77% against 61% for the trained model on 119 unseen parts from a newer batch, at under one cent per part.

Projects & Academic Research

IMC Prosperity 4 – Rank 458/18,800+ (13th in France) [*GitHub*]

Team of 3

- Traded a full algorithmic book: market making, basket arbitrage, EMA mean reversion, and delta-hedged options (vol-smile fitting, Black-Scholes vs. Bachelier); authored ~93 of the team's 120 commits.
- Built an Optuna-based strategy optimizer (TPE/CMA-ES) with anti-overfitting validators (Deflated Sharpe Ratio, PBO/CSCV) and a dashboard of 13 market-microstructure panels (order-flow imbalance, mark-outs by counterparty, lead-lag CCF).

Prediction-Market Trading Strategies — Polymarket [*track record*]

Independent (two-person project)

- Priced Polymarket's daily BTC/ETH binaries off Deribit's options chain: Black-76 on the forward, an SVI smile fitted per expiry by least squares in total-variance space (PCHIP fallback on thin slices), and Breeden-Litzenberger on bracketing strikes for the risk-neutral density; a d_2 -space adjustment (Girsanov funding drift, variance risk premium, risk reversal, Cornish-Fisher skewness) mapped it to real-world probabilities.
- Sized with quarter-Kelly under a per-asset delta budget. An earlier generation matched equivalent markets across venues using sentence embeddings and an LLM; later ones added multi-outcome and 15-minute crypto markets, and a 102-feature LightGBM expected-value gate over L2 order-book and on-chain features, walk-forward-validated on 82K+ self-collected snapshots.
- 13 strategies across three generations; \$260k traded over 6,213 markets for +\$155 net. Wrote the postmortems up publicly rather than the flattering numbers: settlement-time leakage that had held a backtest Sharpe at 9.5 when the real figure was 0.85, and why the options edge died once the daily chain's narrow strike coverage forced interpolations the size of the edge.

Technical Skills

Quantitative Finance: Market making, alpha research, signal evaluation (rank IC, Newey-West, Deflated Sharpe/PBO), factor models (Axioma residualization), market microstructure (L2 order books, order-flow imbalance, mark-outs, maker/taker mechanics), derivative pricing (Black-Scholes, Black-76, Bachelier, IV surfaces, delta hedging), Kelly sizing

Programming: Python (advanced) | **Human languages:** Native French and English

Trading Systems: asyncio/aiohttp, WebSocket market data, fail-closed order gateways, pre-trade risk pipelines, event-driven backtest engines, walk-forward validation, Postgres/SQLite

Machine Learning: PyTorch, transformers/attention, LightGBM, XGBoost, Optuna, GNNs, RL (PPO, GRPO), LLM post-training (QLoRA, DPO), HPC (SLURM, A100)

Tools: Git, Linux, Docker, SLURM, LaTeX, Claude Code (agentic workflows, custom skills per repo)

Achievements: 1st, CentraleSupélec GreenAI hackathon | 2nd, HackEurope Solana prize (\$1,000, Feb 2026)